

Adversarial self-review of structural-bounds-framework v0.4

Pre-empt of likely reviewer objections + proposed responses

Rohan Fossé

Gaël Pallares

CESI LINEACT (EA 7527), Montpellier, France

Working notes, 2026-05-24 — *not for distribution beyond co-authors*

Abstract

We enumerate likely objections from a JMLR / FoCM editorial pass and a Gemini-Pro-2.5-style adversarial reviewer. For each, we propose a response and a manuscript edit category (text-only / new theorem / new experiment / acknowledge open question). The intent is to ratchet the manuscript through the most predictable critiques before formal submission.

1 Theoretical objections

[MAJOR] “The lower bound is trivial under expansive hypothesis classes.” A neural network with sufficient capacity has $\mathcal{H}_d = \mathbb{R}^N$, $R_{\text{spec}}^2(\mathcal{H}_d, \mathbf{y}) = 1$, and the bound reduces to $\mathbb{E}[L_{T^c}] \geq 0$. Reviewer may object that the bound has no content in the practically interesting regime of large-capacity models.

Response. (a) The bound’s interest is at fixed $\mathcal{H}_d$, where it characterises the irreducible component of LSO error for any algorithm operating within that class. Real neural networks have implicit biases (small effective $\mathcal{H}_d$) that the bound’s *learner-specific* formulation captures; the empirically operative $\mathcal{H}_d$ is not the parametric capacity but the SGD-reachable subspace (Neyshabur et al., 2015), which is much smaller. (b) The bound is most informative for the regime where $R_{\text{spec}}^2 \in (0, 1)$, i.e., where the learner’s class is a non-trivial subset of $\mathbb{R}^N$. This is the standard regime for materials informatics (Magpie / CHGNet), mobility (IMD-4), and recommender systems (item-feature subspaces). Manuscript edit: add a Remark in §Setup about the trivial-vs-informative regimes.

[MAJOR] “The Berry-Esseen lower bound uses an adversarial signal in $\mathcal{H}_d^\perp$; this is unrealistic for practical signals which are typically $\mathcal{H}_d$ -aligned by design.” Reviewer may object that the lower bound is not informative for the typical regime where the practitioner has chosen $\mathcal{H}_d$ to capture most of $\mathbf{y}$.

Response. The lower bound is exact in expectation regardless of signal-class alignment (Theorem 1). The Berry-Esseen high-probability slack *additionally* forces a worst-case fluctuation; this is not the typical-case slack but the worst-case one. For the typical case, the bound is tight in expectation with no slack. The argument also shows that the algorithm cannot do better than the floor on *adversarial* signals, ruling out claims of universally-tight performance. Manuscript edit: add a paragraph in §Discussion clarifying that the lower bound is informative in two distinct ways (typical-case in-expectation; worst-case high-probability).

[MINOR] “The Rademacher complexity $\mathcal{R}_n^{\text{trans}}(\mathcal{H}_d)$ is not estimable from data.” Reviewer may ask for an explicit upper bound on $\mathcal{R}_n^{\text{trans}}$ for the regressors used in the empirical section (LightGBM, kernel ridge).

Response. For closed-linear $\mathcal{H}_d$ of dimension d , $\mathcal{R}_n^{\text{trans}}(\mathcal{H}_d) \leq C\sqrt{d/u}$ for an absolute constant C (Lemma B.1 of SI). For LightGBM, the effective d is bounded by $T \cdot L \cdot d_{\text{leaves}}$ where T

is the tree count, L the depth, d_{leaves} the per-leaf dimension (usually 1). Manuscript edit: in SI Appendix B, add a paragraph with explicit $\mathcal{R}_n^{\text{trans}}$ bounds for the regressor families appearing in Table 1.

[MINOR] “The negative-transfer spectral-disjointness condition is hard to verify in practice.” How does a practitioner know whether their source / target are spectrally disjoint?

Response. The condition is checkable by computing the joint graph G on the union $V_{\text{src}} \cup V_{\text{tgt}}$, then computing the projection of $\mathbf{y}_{\text{src}}$ and $\mathbf{y}_{\text{tgt}}$ onto the bottom- K Laplacian eigenvectors. For closed-linear $\mathcal{H}_d$, the relevant projection is $\mathbf{P}_{\mathcal{H}_d} \mathbf{y}$, computable in $O(d^2 N)$ time. This is the same computational cost as the spectral R^2 computation already used in Corollary C1 empirically. Manuscript edit: add a paragraph in §C2 with the practical computational recipe.

[NIT] “The factor- $\sqrt{\rho - 1}$ improvement from transductive Rademacher (§notes/04) is undersold.” Reviewer may note that this is a non-trivial result on its own.

Response. Agreed; notes/04 stands alone as a methodological contribution. Manuscript edit: elevate the transductive Rademacher observation from a passing Remark in §6 to a labelled paragraph in the Contributions list (§1), making explicit that this is a substantive improvement over the naive Mohri-routed proof.

2 Empirical objections

[MAJOR] “Spearman $\rho = +1.000$ on $n = 8$ tasks is suspiciously perfect.” Reviewer may question whether the rank-order match is too good and indicates over-fitting to the framework.

Response. (a) The exact permutation test gives $p = 4.96 \times 10^{-5}$, accounting for the small- n over-claim risk. (b) The partial Spearman controlling for $\log N$ and $\log \text{Var}(\mathbf{y})$ is $+0.79$ with $p = 0.032$, confirming the signal survives the obvious confounders. (c) The shuffled- k NN null on a degree-and-clustering-matched graph gives partial $\rho = +0.25$, $p = 0.31$, showing the chemistry graph is essential, not the GroupKFold protocol per se. (d) The CIG ratio of $18\times$ to $145\times$ above null further rules out accidental alignment. No manuscript edit needed; falsifiability section already addresses this.

[MINOR] “The cross-domain pilots (MovieLens, QSAR, AFLOW) lack R_{spec}^2 values, breaking the consistency of Table 1.” Reviewer may ask why these rows have “—” in the R_{spec}^2 column.

Response. Computing R_{spec}^2 requires a domain-specific similarity graph, which is non-trivial outside materials informatics (QSAR molecules need pharmacophore-similarity graphs; MovieLens users need user-user similarity from collaborative-filtering structure). We deliberately chose to report only the ΔR_{LSO}^2 values for these rows to avoid hand-tuning graph choices for the cross-domain pilots. Manuscript edit: add a sentence in §sec:empirics explaining the deliberate omission.

[MINOR] “Bike-share and mobility rows have ranges or are unfilled; not a serious empirical anchor.”

Response. The bike-share row reports $\Delta R^2 \in [+0.06, +0.55]$ across 9 networks (from bikeshare-demand-for README, d19 leave-station-out). The mobility row is from a companion paper in preparation; we report a placeholder rather than omit, with explicit notation. Both are secondary anchors; the primary anchor is the 8-task MatBench panel with its falsifiability guarantees. Manuscript edit: replace the mobility “—” values with a concrete summary statistic once mobility-applicability-bound v0.2 is ready.

[NIT] “Figure 1’s marker-size encoding for $\log_{10} N$ is hard to read.”

Response. Agreed; a sub-figure with a marker-size legend would help. Manuscript edit (post-v1.0): add a small inset legend showing marker scale for $N = 10^3, 10^4, 10^5$.

3 Methodological objections

[MAJOR] “Why not just cite the existing leave-one-out PAC-Bayes literature?” Reviewer (likely from the PAC-Bayes lane) may ask why our framework is needed when leave-one-out PAC-Bayes (Langford and Caruana, 2002; Dziugaite and Roy, 2017) already exists.

Response. PAC-Bayes bounds are upper bounds on excess risk under *stochastic* predictors; ours is a lower bound on *any* deterministic ERM under leave-node-out. The two lanes are complementary and largely non-overlapping; our framework provides the irreducible floor that PAC-Bayes upper bounds aspire toward. Manuscript edit: add a paragraph in §1.2 (Relation to prior work) explicitly distinguishing the two lanes.

[MINOR] “The Cramér-Rao analogy is loose — there is no Fisher information here.”

Response. The analogy is structural, not parametric. Cramér-Rao bounds the variance of any unbiased estimator from below by $1/I(\theta)$; we bound the expected leave-node-out loss from below by the irreducible $(1 - R_{\text{spec}}^2)\text{Var}(\mathbf{y})$. Both establish that the estimator’s reach (parametric for Cramér-Rao, geometric for ours) fixes a floor that no algorithmic ingenuity can beat. We do not claim a literal Fisher-information identity; the analogy is at the level of the *role* of the irreducibility floor. Manuscript edit: soften the Cramér-Rao language in §1 and §4 if reviewer pushback is substantial; alternatively, keep the analogy but add a Remark clarifying what is and is not analogous.

[NIT] “The transductive Rademacher constant 5.05 is mysterious.”

Response. The constant decomposes as $11/4 + \sqrt{\log(2/\eta)}/2$ for $\eta = 0.05$ (SI Appendix B, around eq. for $\mathcal{R}_n^{\text{trans}}$). Manuscript edit: expand the SI derivation to spell this out explicitly.

4 Editorial objections

[MINOR] “Length is borderline excessive for a JMLR submission (20 pp main + 5 pp SI + 5 notes + cover letter).”

Response. JMLR has no hard page limit and routinely publishes 20–40 pp manuscripts; the SI compresses additional technical content that would otherwise inflate the main paper. Notes are research-stage documents, not part of the submission package per se — only the main paper + SI go to the editorial portal. No manuscript edit needed.

[MINOR] “Why JMLR and not Annals of Statistics or IEEE TIT?”

Response. JMLR is the natural home for this work because it unifies a theoretical contribution (the bound) with a substantial empirical-validation programme (8 MatBench tasks + cross-domain pilots); Annals of Statistics and TIT lean too theoretical for the empirical companion. FoCM is the explicit backup. Manuscript edit: no change; cover letter already addresses this.

[NIT] “Bibliography is not author-year-consistent (some entries lack DOIs).”

Response. Most entries have DOIs; a few historical ones (Bikelis 1969, Talagrand 1996) lack them as the journals were not DOI-indexed at publication. No manuscript edit needed.

5 Triage plan toward v0.5

Edits needed for the next manuscript milestone (v0.5), in priority order:

P1. Practical recipe paragraph in §C2 (NT) for the spectral-disjointness check. $O(d^2N)$ computation; 1 paragraph.

P2. Add Remark in §Setup on the trivial-vs-informative regimes for varying $\mathcal{H}_d$ capacity.

- P3. Elevate the transductive Rademacher contribution** from a Remark to a labelled item in the Contributions list (§1).
- P4. Add deliberate-omission sentence** in §sec:empirics about cross-domain R_{spec}^2 values being out of scope.
- P5. Add a paragraph in §1.2** distinguishing the leave-one-out PAC-Bayes lane from our spectral lane.
- P6. Add a paragraph in §Discussion** clarifying the two distinct senses in which the lower bound is informative (typical / worst-case).
- P7. Expand SI Appendix B** with explicit transductive Rademacher bounds for LightGBM and kernel ridge, plus derivation of the 5.05 constant.

None of these are blockers for v0.4 $\rightarrow$ v0.5; all are text-only edits that can be done in a focused 2-day session.

Acknowledgements (working notes)

These notes pre-empt likely reviewer objections by enumerating them explicitly. The convention is borrowed from the **materials-applicability-bound** repository, which incorporated Gemini Pro 2.5 reviewer comments in two rounds before MLST submission. The present notes are intended for a single Gemini Pro 2.5 self-review pass before tagging v0.5.

References

- Gintare Karolina Dziugaite and Daniel M. Roy. Computing nonvacuous generalization bounds for deep (stochastic) neural networks with many more parameters than training data. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2017. arXiv:1703.11008.
- John Langford and Rich Caruana. (not) bounding the true error. *Advances in Neural Information Processing Systems (NIPS)*, 2002.
- Behnam Neyshabur, Ryota Tomioka, and Nathan Srebro. In search of the real inductive bias: On the role of implicit regularization in deep learning. In *International Conference on Learning Representations (ICLR) Workshop*, 2015. arXiv:1412.6614.